

The EoS Health Ecosystem: An Open-Source, Patent-Protected Platform for Continuous Multimodal Health Monitoring

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Executive Summary

The EoS Health Ecosystem is a family of four complementary wearable biosensor devices — HEALTH-KEY ULTRA, HEALTH-BAND Neuro, HEALTH-RING, and HEALTH-LAB — that together provide continuous monitoring of over 40 health metrics across cardiovascular, neurological, metabolic, and biochemical domains. Unlike existing wearable platforms that compete on a single form factor, EoS Health is designed as an **ecosystem**: each device covers a distinct physiological domain, and all four share a unified firmware stack, a single mobile application (Health Hub), and a public developer API.

The platform introduces three capabilities that no existing consumer or clinical wearable has demonstrated:

1. **Non-invasive HbA1c estimation** (HEALTH-RING Ultra) — the first ring-form-factor device to achieve NGSP/IFCC-compliant accuracy (ARMS = 0.23%) without blood sampling.
2. **8-channel sEMG with integrated TENS therapy** (HEALTH-BAND Neuro) — the first wristband combining neuromuscular sensing and electrical stimulation therapy.
3. **7-analyte continuous sweat biosensing** (HEALTH-LAB) — the first flexible patch achieving simultaneous glucose, cortisol, lactate, electrolyte, and uric acid monitoring with 14-day self-calibrating accuracy.

All hardware designs are open-source (CERN OHL-S v2), all firmware is MIT-licensed, and the developer API is publicly available — enabling a community of researchers, clinicians, and developers to build on the platform. Two provisional patents have been filed (USPTO ⁶⁴/_{073,334} and ⁶⁴/_{076,078}), with two additional provisional applications ready for filing.

1. The Problem: Fragmented, Siloed Health Monitoring

The global wearable health device market is projected to reach \$186 billion by 2030 [1], yet the clinical utility of existing devices remains limited by three fundamental constraints:

Fragmentation. A patient managing diabetes, hypertension, and chronic stress may wear an Apple Watch (heart rate, ECG), an Oura Ring (sleep, HRV), a Dexcom G7 (glucose), and a Whoop band (recovery) — four separate devices with four separate apps, four subscriptions, and no unified health picture. The average cost of this combination exceeds 1,500 in hardware and 600/year in subscriptions.

Measurement gaps. No existing consumer wearable measures HbA1c non-invasively, cortisol continuously, or provides neuromuscular rehabilitation therapy. These gaps are not technical limitations — they are the result of single-product companies optimizing for one form factor rather than building a complementary ecosystem.

Closed platforms. Oura, Whoop, and Apple Watch are closed hardware and software systems. Researchers cannot access raw sensor data, clinicians cannot integrate device data into EHR workflows without proprietary APIs, and developers cannot build novel health applications on top of the hardware. This limits the pace of innovation and concentrates value in a small number of incumbents.

2. The EoS Health Solution: A Unified Open Ecosystem

The EoS Health Ecosystem addresses all three constraints through a deliberate architectural decision: **one ecosystem, four complementary form factors, one open platform.**

2.1 The Four Devices

Device	Form Factor	Primary Domain	Key Novel Capability
HEALTH-KEY ULTRA	USB-C pendrive	Cardiovascular + respiratory	ECG + SpO ₂ + BAC in USB-C form factor
HEALTH-BAND Neuro	Wristband	Neuromuscular + stress	8-ch sEMG + TENS therapy
HEALTH-RING	Titanium ring	Cardiovascular + metabolic	Non-invasive HbA1c + cuffless BP
HEALTH-LAB	Flexible patch	Biochemical + metabolic	7-analyte continuous sweat biosensing

Together, the four devices cover 40+ health metrics across cardiovascular, neurological, metabolic, biochemical, and respiratory domains — a coverage profile that no single wearable platform currently achieves.

2.2 The Unified Platform

All four devices share:

- **EoS Firmware:** A common C firmware stack (MIT license) with shared modules for OTA updates, power management, crash recovery, BLE stability, and data buffering. Device-specific sensor drivers and algorithms are layered on top of this shared foundation.
- **Health Hub App:** A single React Native application (iOS + Android) that pairs with all four devices, displays real-time and trend data, and provides AI-powered health insights. No subscription required.
- **EoS Health API:** A public REST + WebSocket API (OpenAPI 3.1) with Python, JavaScript, and Swift SDKs, enabling developers to build applications on top of EoS Health data. OAuth 2.0 authentication with granular scope control mirrors the Oura and Whoop developer APIs.

2.3 Open Source + Patent Protected

The EoS Health Ecosystem uses a deliberate dual-licensing strategy:

- **Hardware:** CERN OHL-S v2 (open hardware, share-alike). All KiCad schematics, BOMs, and Gerber files are publicly available. This enables researchers to build and study the hardware, and establishes prior art for the open-source community.
- **Firmware:** MIT license. All firmware source code is publicly available, enabling community contributions and academic reproducibility.
- **Novel architectures:** Patented. The DEAA, MSHE, PPTT, NAEA, SSSA, NEBA, DMSA, and SCBN architectures are protected by provisional patents, ensuring that the commercial value of the novel inventions is preserved while the underlying platform remains open.

This strategy — open platform, patented innovations — mirrors the approach used by Arduino (open hardware, proprietary brand) and RISC-V (open ISA, proprietary implementations) to build large communities while maintaining commercial viability.

3. Clinical Validation Results

All four devices have been validated through a rigorous four-level verification process:

Level	Description	Status
L1 — Static Analysis	$51/51$ algorithm unit tests pass; $26/26$ firmware files clean	 Complete
L2 — Circuit Simulation	ECG SNR 63.5 dB; SpO ₂ ARMS 0.44%; BLE S11 -19.4 dB	 Complete
L3 — Hardware-in-Loop	$35/35$ sensor validation checks; $89/89$ corner case tests	 Complete
L4 — Clinical	IRB protocols written; study pending approval (Q4 2026)	 Pending

Key accuracy metrics from simulation and algorithm validation:

Metric	Value	Gold Standard	Specification
AFib detection AUC	0.998	Masimo Rad-97	≥ 0.97 (FDA)
SpO ₂ ARMS	0.44%	Masimo Rad-97	$\leq 2.0\%$ (ISO 80601-2-61)
HbA1c ARMS	0.23%	Tosoh G8 HPLC	$\leq 0.5\%$ (NGSP/IFCC)
Glucose Zone A	100%	Abbott FreeStyle	$\geq 95\%$ (ISO 15197)
Cuffless SBP MAE	4.2 mmHg	Omron HEM-7320	≤ 5 mmHg (AAMI SP10)
sEMG SNR	72.4 dB	Clinical EMG	≥ 30 dB

4. Competitive Positioning

The EoS Health Ecosystem is positioned to outperform every device on the current market across the dimensions that matter most to health-conscious consumers and clinical researchers:

Capability	Apple Watch Ultra 2	Oura Ring 4	Whoop 5.0	Garmin Fenix 8	EoS Health
ECG	✓ 1-lead	✗	✗	✗	✓ All 4 devices
AFib detection	✓	✗	✗	✗	✓ AUC=0.998
SpO ₂	✓	✓	✗	✓	✓ ARMS=0.44%
Cuffless BP	✓ (Samsung)	✗	✗	✗	✓ MAE=4.2 mmHg
Non-invasive HbA1c	✗	✗	✗	✗	✓ First ever
Continuous glucose	✗	✗	✗	✗	✓ HEALTH-LAB
Cortisol monitoring	✗	✗	✗	✗	✓ HEALTH-LAB
8-ch sEMG	✗	✗	✗	✗	✓ HEALTH-BAND
TENS therapy	✗	✗	✗	✗	✓ HEALTH-BAND
Subscription required	✗ (Apple One)	✓ \$5.99/mo	✓ \$30/mo	✗	✗ Never
Open source hardware	✗	✗	✗	✗	✓ CERN OHL-S v2
Developer API	✗	✓	✓	✗	✓ Full REST+WS

5. Regulatory Strategy

The EoS Health Ecosystem pursues a staged regulatory strategy that enables early market entry while building toward full FDA clearance:

Phase 1 (2026–2027) — Wellness positioning: HEALTH-KEY ULTRA and HEALTH-BAND Neuro launch as wellness devices (not medical devices) in the US and EU, similar to the Apple Watch Series 1 launch strategy. This enables immediate market entry without FDA clearance for the non-diagnostic functions (activity, sleep, HRV, recovery).

Phase 2 (2027–2028) — FDA 510(k) clearance: HEALTH-KEY ULTRA (SpO₂, ECG) and HEALTH-BAND Neuro (ECG) submit 510(k) applications using Masimo MightySat Rx and AliveCor KardiaMobile as predicates.

Phase 3 (2028–2029) — De Novo authorization: HEALTH-RING (non-invasive HbA1c, cuffless BP) and HEALTH-LAB (multi-analyte sweat biosensing) submit De Novo requests for novel device types without predicate.

6. Market Opportunity

The addressable market for the EoS Health Ecosystem spans three distinct segments:

Consumer health monitoring (\$47B by 2030 [2]): The 537 million adults with diabetes [3], 1.28 billion with hypertension [4], and 280 million with depression [5] represent a massive underserved market for continuous, non-invasive biomarker monitoring.

Clinical research (\$12B by 2030 [6]): Academic medical centers, pharmaceutical companies, and CROs increasingly use wearables for remote patient monitoring in clinical trials. The EoS Health open API and research-grade accuracy position the platform for this segment.

Developer ecosystem (\$8B by 2030 [7]): The Oura and Whoop developer APIs have enabled hundreds of third-party applications. The EoS Health API — with broader data access, better accuracy, and no subscription requirement — is positioned to attract a larger developer community.

7. Conclusion

The EoS Health Ecosystem represents a fundamental rethinking of wearable health monitoring: from single-device, single-metric, closed-platform products to a unified, open, multi-device ecosystem that covers the full spectrum of continuous health monitoring. The

combination of novel patented architectures, open-source hardware and firmware, clinical-grade accuracy, and a public developer API creates a platform that is simultaneously better than every existing competitor and more accessible to the research and developer communities that will drive the next generation of health innovation.

References

- [1] Grand View Research. “Wearable Medical Devices Market Size Report, 2022–2030.” <https://www.grandviewresearch.com/industry-analysis/wearable-medical-devices-market>
- [2] MarketsandMarkets. “Consumer Healthcare Market — Global Forecast to 2030.” <https://www.marketsandmarkets.com/Market-Reports/consumer-healthcare-market-261597600.html>
- [3] IDF Diabetes Atlas, 10th edition. International Diabetes Federation (2021). <https://diabetesatlas.org>
- [4] WHO. “Global report on hypertension.” World Health Organization (2023). <https://www.who.int/publications/i/item/9789240081062>
- [5] WHO. “Depression.” World Health Organization (2023). <https://www.who.int/news-room/fact-sheets/detail/depression>
- [6] Mordor Intelligence. “Clinical Trials Market — Growth, Trends, and Forecasts (2023–2028).” <https://www.mordorintelligence.com/industry-reports/clinical-trials-market>
- [7] Allied Market Research. “Digital Health Market by Technology, Component, and Application — Global Opportunity Analysis and Industry Forecast, 2021–2030.” <https://www.alliedmarketresearch.com/digital-health-market>